Experiment:14

TCL COMMANDS – COMMIT, SAVEPOINT, ROLLBACK

AIM:

To learn how to use various TCL commands Commit, Savepoint and Rollback SQL commands

Procedure and Syntax:

Transaction Control Language (TCL) commands are used to manage transactions in the database. These are used to manage the changes made to the data in a table by DML statements. It also allows statements to be grouped together into logical transactions.

COMMIT:

• COMMIT command is used to permanently save any transaction into the database.

• When we use any DML command like INSERT, UPDATE or DELETE, the changes made by these commands are not permanent, until the current session is closed, the changes made by these commands can be rolled back.

• To avoid that, we use the COMMIT command to mark the changes as permanent

Syntax:

COMMIT;

ROLLBACK:

➢ This command restores the database to last committed state.

➢ It is also used with SAVEPOINT command to jump to a savepoint in an ongoing transaction.

➢ If we have used the UPDATE command to make some changes into the database, and realize that those

changes were not required, then we can use the ROLLBACK command to rollback those changes, if they were not committed using the COMMIT command.

Syntax:

ROLLBACK;

ROLLBACK TO savepoint_name;

SAVEPOINT:

➢ SAVEPOINT command is used to temporarily save a transaction so that you can rollback to that point whenever required.

Syntax:

SAVEPOINT savepoint_name;

OUTPUT:

Experiment:14

```
mysql> CREATE TABLE class (
    ->     name VARCHAR(30),
    ->     id INT
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO class VALUES ('alpha', 1);
Query OK, 1 row affected (0.05 sec)

mysql> INSERT INTO class VALUES ('beta', 2);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO class VALUES ('charlie', 3);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO class VALUES ('delta', 4);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO class VALUES ('echo', 5);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM class;
+---------+------+
| name    | id   |
+---------+------+
| alpha   |    1 |
| beta    |    2 |
| charlie |    3 |
| delta   |    4 |
| echo    |    5 |
+---------+------+
5 rows in set (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE class SET name = 'bravo' WHERE id = 5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SAVEPOINT A;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO class VALUES ('uppal', 6);
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT B;
```

```
mysql> INSERT INTO class VALUES ('balu', 7);
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT C;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM class;
+---------+------+
| name    | id   |
+---------+------+
| alpha   |    1 |
| beta    |    2 |
| charlie |    3 |
| delta   |    4 |
| bravo   |    5 |
| uppal   |    6 |
| balu    |    7 |
+---------+------+
7 rows in set (0.00 sec)

mysql> ROLLBACK TO B;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM class;
+---------+------+
| name    | id   |
+---------+------+
| alpha   |    1 |
| beta    |    2 |
| charlie |    3 |
| delta   |    4 |
| bravo   |    5 |
| uppal   |    6 |
+---------+------+
6 rows in set (0.00 sec)

mysql> ROLLBACK TO A;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM class;
+---------+------+
| name    | id   |
+---------+------+
| alpha   |    1 |
| beta    |    2 |
| charlie |    3 |
| delta   |    4 |
| bravo   |    5 |
+---------+------+
5 rows in set (0.00 sec)

mysql> commit;
Query OK, 0 rows affected (0.06 sec)

mysql> SELECT * FROM class;
+---------+------+
| name    | id   |
+---------+------+
| alpha   |    1 |
| beta    |    2 |
| charlie |    3 |
| delta   |    4 |
| bravo   |    5 |
+---------+------+
5 rows in set (0.00 sec)
```